Non-Malignant Hematology Protocol Complexity & Workload Rubric (Draft v0.1)

Purpose. Two linked instruments: **Part A** scores intrinsic protocol complexity once (at feasibility, re-scored on amendment). **Part B** converts that score plus live accrual into a monthly workload figure for staffing decisions. They are deliberately separate — a very complex protocol with two enrolled participants generates less work than a simple registry with 400.

Why not just use ASCO/OPAL. Both are anchored in oncology treatment trials and patient-facing encounters. This draft adds weight for the three things that dominate a non-malignant hematology portfolio and are absent or underweighted in those tools: long-horizon follow-up obligations (gene therapy LTFU), observational/registry data burden, and qualitative/community-engaged components.

> **Status: unvalidated draft.** Weights below are reasoned starting points, not empirically derived. Section 5 specifies the calibration work required before this drives any staffing or budget decision.

PART A — Protocol Complexity Score (100 points)

Score each item once. Sum domains for a total. Anchor descriptors are given for the endpoints; scorers interpolate.

Domain 1 — Regulatory & Sponsorship Framework (0–18)

Item	0	Mid	Max	Pts
Regulatory status	Not FDA-regulated	IND/IDE exempt determination required	Active IND/IDE held by St. Jude	0–5
Sponsor type	Investigator-initiated, internal	NIH/federal	Industry, or federal with external coordinating center	0–3
Site role	Single-site	Participating site, multi-site	**Coordinating center** for multi-site	0–5
Oversight bodies	IRB only	+ DSMB or independent monitor	+ external audit/inspection readiness obligations	0–3
IRB model	Local IRB only	sIRB relying	sIRB **reviewing** for external sites	0–2

Domain 2 — Population & Consent (0–12)

Item	Range	Notes
Pediatric enrollment with assent tiers	0–3	0 = adults only; 3 = multiple assent age bands with re-consent at transition
Vulnerable/protected populations	0–3	decisionally impaired, wards, pregnant participants
Re-consent burden	0–3	anticipated amendments requiring re-consent of active cohort
Language/literacy accommodation	0–3	translated materials, interpreter-dependent visits

Domain 3 — Visit & Procedure Burden (0–18)

Item	Range	Notes
Protocol-required visits per participant-year	0–5	0 = ≤1; 5 = >12
Procedure intensity	0–5	0 = none/records only; 5 = apheresis, conditioning/HSCT, sedated imaging
Timed/PK or serial sampling	0–3	any window <30 min counts as max
Visit window rigidity	0–2	±14 d vs ±2 d
Off-site, home, or remote assessment	0–3	includes wearables and home phlebotomy logistics

Domain 4 — Investigational Product / Intervention (0–12)

Item	Range	Notes
IP handling & accountability	0–4	0 = no IP; 4 = cell/gene product with chain-of-custody
Blinding & randomization	0–3	0 = open-label/none; 3 = double-blind with unblinding procedures
Pharmacy/manufacturing interface	0–3	0 = none; 3 = on-site manufacturing or apheresis-to-product workflow
Dose modification complexity	0–2	rules requiring per-visit adjudication

Domain 5 — Data & Specimen Management (0–16)

Item	Range	Notes
eCRF volume	0–4	fields per participant-visit
Number of distinct data systems	0–4	sponsor EDC + REDCap + registry + imaging portal each count
Biospecimen processing/shipping	0–3	0 = none; 3 = time-critical processing with international shipment
Central/core lab or imaging read	0–2	
PRO/ePRO administration	0–2	
Linked or derived data (geospatial, EHR extract, genomic)	0–3	includes SDOH/neighborhood-level linkage

(Domain 5 max is 18 as itemized; cap the domain contribution at 16.)

Domain 6 — Safety Reporting & Monitoring (0–12)

Item	Range	Notes
Expected AE/SAE volume	0–4	
Expedited reporting obligations	0–3	multi-recipient (FDA, sponsor, sIRB, DSMB) = max
Monitoring visit frequency	0–3	
Long-term follow-up horizon	0–2	0 = ⩽2 yr; 2 = ⩾5 yr (gene therapy LTFU obligations)

Domain 7 — Qualitative, Mixed-Method & Community Engagement (0–12)

Item	Range	Notes
Interview/focus group data collection	0–4	count × duration; includes scheduling burden
Transcription, coding & analysis load	0–4	multi-coder with reliability requirements = max
Community advisory board or partner site coordination	0–2	
Dissemination obligations beyond manuscript	0–2	community reports, policy briefs, stakeholder convenings

Complexity tiers

Total	Tier	Label
⩽ 20	1	Minimal
21–38	2	Low
39–58	3	Moderate
59–76	4	High
⩾ 77	5	Very high

PART B — Monthly Workload Index

Monthly Workload Units (WU) for a protocol =

`Static WU (by tier) + Σ (participants in status × per-participant WU) × Phase multiplier`

Static load — accrues whether or not anyone is enrolled

Tier	Static WU/month
1	2
2	4
3	8
4	14
5	22

Per-participant WU per month, by status and tier

Status	T1	T2	T3	T4	T5
In screening / consent	0.5	1.0	1.5	2.5	3.5
Active intervention / on-treatment	1.0	2.0	3.5	6.0	9.0
Active follow-up (protocol visits)	0.5	1.0	1.5	2.5	3.5
Long-term follow-up (annual contact)	0.1	0.2	0.3	0.5	0.7
Closed to accrual, data cleaning/closeout	0.2	0.3	0.5	0.8	1.0

Phase multipliers — apply to the protocol total

Condition	Multiplier
Startup (activation −3 mo to first enrollment)	1.6
Steady state	1.0
Substantive amendment month (+ following month)	1.3
Audit/inspection or monitoring visit month	1.4
Closeout quarter	1.2

Converting WU to FTE

Do **not** adopt a national ratio. Establish your own capacity constant C = WU one 1.0 FTE coordinator sustains per month, via the time study in Section 5. Then:

`Required CRC FTE = Total portfolio WU/month ÷ C`

For reference on order of magnitude: a published oncology instrument (IWAT) settled on a monthly score of 500–600 as appropriate for a full-time CRC, on a different scale from this one. Treat that as evidence that the calibration step matters, not as a number to import.

Two corrections to apply before acting on the output:

1. **Non-substitutability.** A regulatory coordinator's WU and a bedside CRC's WU are not interchangeable. Compute portfolio WU separately by role before converting to FTE, or you will "solve" a regulatory bottleneck by hiring a nurse.
2. **Concurrency penalty.** Coordinators carrying >5 protocols lose time to context-switching that this model does not capture. Apply a 1.1 multiplier at 6–8 protocols, 1.2 at 9+, until your own data says otherwise.

PART C — Governance

- **Score at feasibility review**, before institutional commitment. The score is an input to the go/no-go, not a post-hoc justification.
- **Two independent scorers**, reconciled. Log both raw scores — you need them for reliability testing.
- **Re-score** annually and on any amendment touching Domains 3, 4, 5, or 6.
- **Freeze weights for 12 months** once calibrated. Mid-cycle weight changes destroy the trend data that makes this useful for budget requests.

PART D — Build notes (REDCap)

- One instrument for Part A (scored once per protocol, versioned via repeating instrument so amendment history is preserved), one repeating instrument for monthly Part B counts.
- Put the domain sums and tier assignment in `@CALCTEXT` / calculated fields so scorers never do arithmetic.
- Pull the Part B status counts from your CTMS if it can produce them; hand entry across 50+ protocols monthly will not survive contact with reality.
- Build the reliability export (both scorers' raw item scores, wide format) on day one, not after you need it.

PART E — Validation plan

Given the intended use — staffing and budget justification — this needs more than face validity.

1. **Content validity.** Panel of 6–8 (CRC, regulatory coordinator, data manager, nurse, PI, finance) rates each item for relevance; compute I-CVI/S-CVI, drop items <0.78.

2. **Inter-rater reliability.** 20–25 protocols spanning all five tiers, two independent scorers. Target weighted $\kappa \geq 0.70$ at item level, ICC ≥ 0.80 on total score. Items failing this need better anchors, not removal.
3. **Criterion validity / capacity constant.** 6–8 week prospective time study on a stratified sample of staff. Regress actual hours on predicted WU. The slope gives you `C`; the residuals tell you which domains are mis-weighted.
4. **Known-groups check.** Scores should separate protocols staff already identify as heavy vs. light. If they don't, the instrument is wrong and the staff are right.
5. **Re-calibration** every 24 months, or after any material change in portfolio composition.

Steps 1–3 constitute a publishable methods paper — there is a documented gap for complexity instruments outside oncology, and a non-malignant hematology instrument covering interventional, observational, and qualitative designs would fill it.